Blockchain

Slides by Andrew DeOrio and Mike Cafarella, with some slides from Joseph Bonneau and Eric Wustrow.
Some technical content from “The Economics of Bitcoin, or Bitcoin in the Presence of Adversaries”.

Blockchain

- *Blockchain* is a shared database with no central owner. Everyone keeps a copy and updates requires agreement.
- Many uses for a blockchain
- Bitcoin cryptocurrency was the first
- https://www.youtube.com/watch?v=yubzJw0uiE4

Agenda

- **Background: currency crash course**
- Bitcoin
- Blockchain
- Mining
- Altcoins
- Summary

Traditional currency

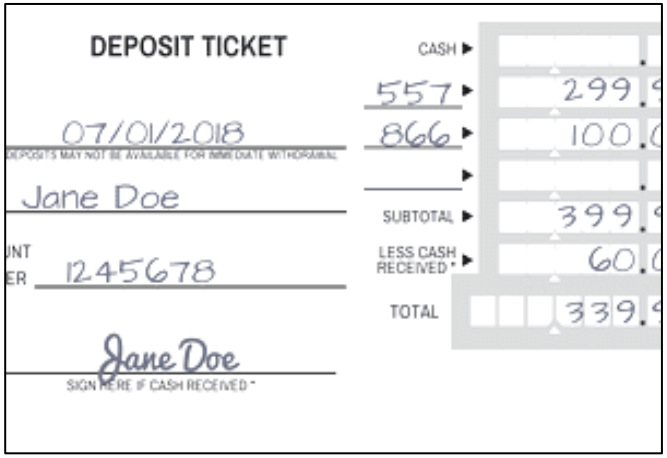
- What is money?
  - *Currency* is a medium of exchange
- A convenient way to buy and sell stuff
  - Easier than bartering "I'll fix your laptop if you give me one dozen eggs."
  - Easier than carting around Gold bullion

Fiat currency

- *Fiat money* is a currency without intrinsic value
- Often created by government regulation
- It's valuable because two parties agree it has value

Digital currency

- *Digital currency* is currency available in digital form
- Example: Bank deposit
  - After deposit, the money exists only as a number on the bank's balance sheet
  - You can spend it by transferring it, e.g., to pay a credit card bill



Central bank

- In the U.S., the central bank is the Federal Reserve
  - "Fed" for short
- Decides how much money exists, e.g., when to print money
- Bank used by banks
  - Lends money to banks
- Another digital currency example: central bank reserve
  - Money exists as a number on the central bank's balance sheet

Central bank

- A central bank is centrally controlled
- Example: the U.S. controls how much U.S. Dollars are worth AKA how many there are
- If the Fed wants more dollars, the Fed can print more
  - Print more dollars. Don't even need to print them. Just increase the number on the balance sheet.
  - *Quantitative easing*

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Bitcoin

- First decentralized digital currency
- First cryptocurrency
- No central authority
- Solves “double spending” problem
- Uses cryptography and distributed systems

Bitcoin

Bitcoin is also ...

- A get-rich-quick scheme
- A fiat currency without a government behind it
- A handy resource for criminals
- The future of money… maybe?

Satoshi Nakamoto

- Created in 2008 under pseudonym Satoshi Nakamoto
- *2008: Nakamoto posted: "Bitcoin: A Peer-to-Peer Electronic Cash System"* to a cryptography mailing list
- Code released 2009
- Identity unknown
- 2010 Nakamoto disappears after mining about 1,000,000 BTC

Gov't backed currency vs. Bitcoin

Government backed

- Analog and digital
- Fiat currency
- Centralized
  - Controlled by a central bank
  - Represented by a single bank ledger

Bitcoin

- Digital
- Fiat currency
- Decentralized
  - Controlled by a distributed algorithm
  - Represented by shared ledger

Bitcoin basics

- A bitcoin is a digital asset
- A Bitcoin address is a public key (AKA Bitcoin Wallet)
- Whoever has the matching private key controls that bitcoin
- To send bitcoin, the owner signs a transaction transferring it to another address
- Ownership is tracked through a chain of digitally signed transactions

Spending Bitcoin

- Alice pays Bob
- Alice signs transaction and broadcasts it to all Bitcoin nodes
- All nodes must agree on a sequence of transactions
- Added to Blockchain

- Sequence is *Blockchain*
- Process of agreement is *Bitcoin mining*

signed by Alice
Pay to pk_{Bob} : H( )

Bitcoin and anonymity

- Bitcoin is **NOT** anonymous
- Everyone can see every Bitcoin transaction
  - It's a distributed public ledger
- No one can see who owns each public key
  - Unless you tell them
- This is called pseudo anonymous

Blockchain tracing

- If I find out John Doe's public key, what can I learn?

- If I know the Bitcoin wallet public key of a Darknet market vendor selling illegal goods, what can I learn?

Blockchain tracing

- If I find out John Doe's public key, what can I learn?
  - Every transaction he's ever done
    - Other party's Bitcoin wallet public key
  - Bitcoin public key for markets are well known
    - E.g., if he bought something from a Darknet Market
- If I know the Bitcoin wallet public key of a Darknet market vendor selling illegal goods, what can I learn?
  - All their customers, ever
    - Other party's Bitcoin wallet public key
- *Blockchain tracing* AKA *blockchain analysis*

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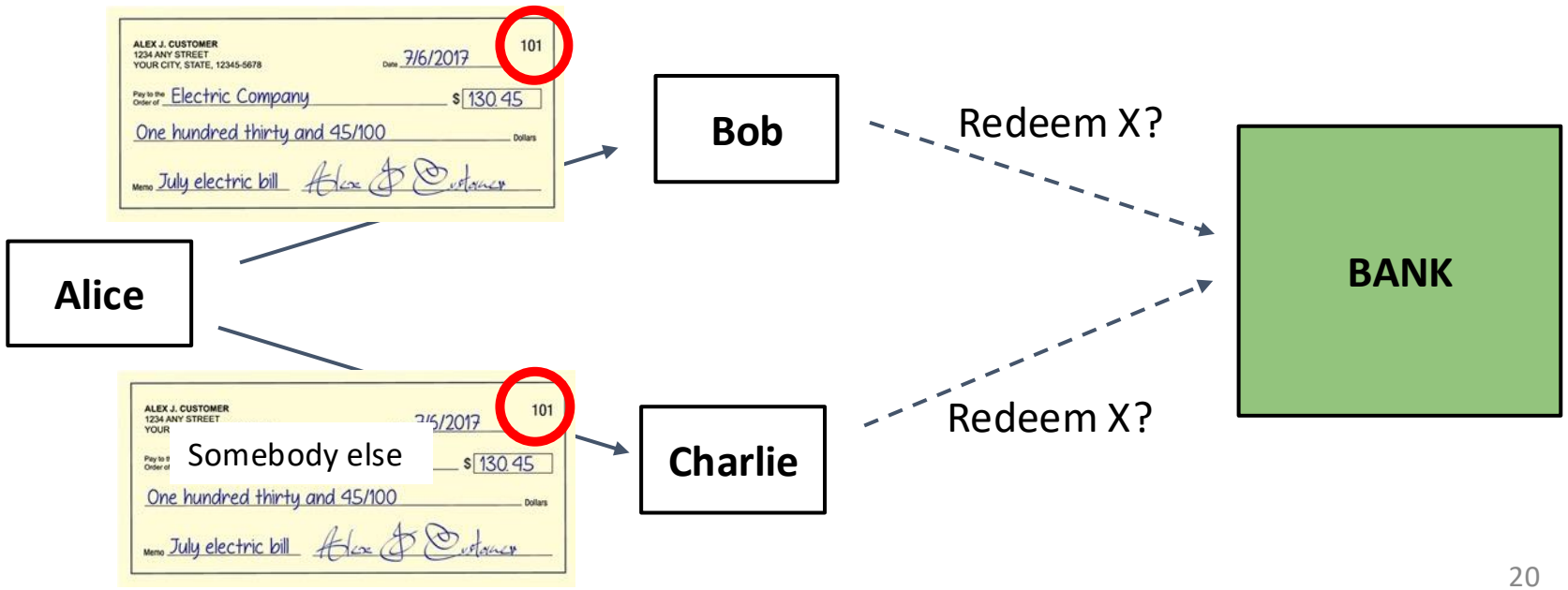
Double spending

- Double spending: Same money spent twice.
- In US, ACH system prevents double spending

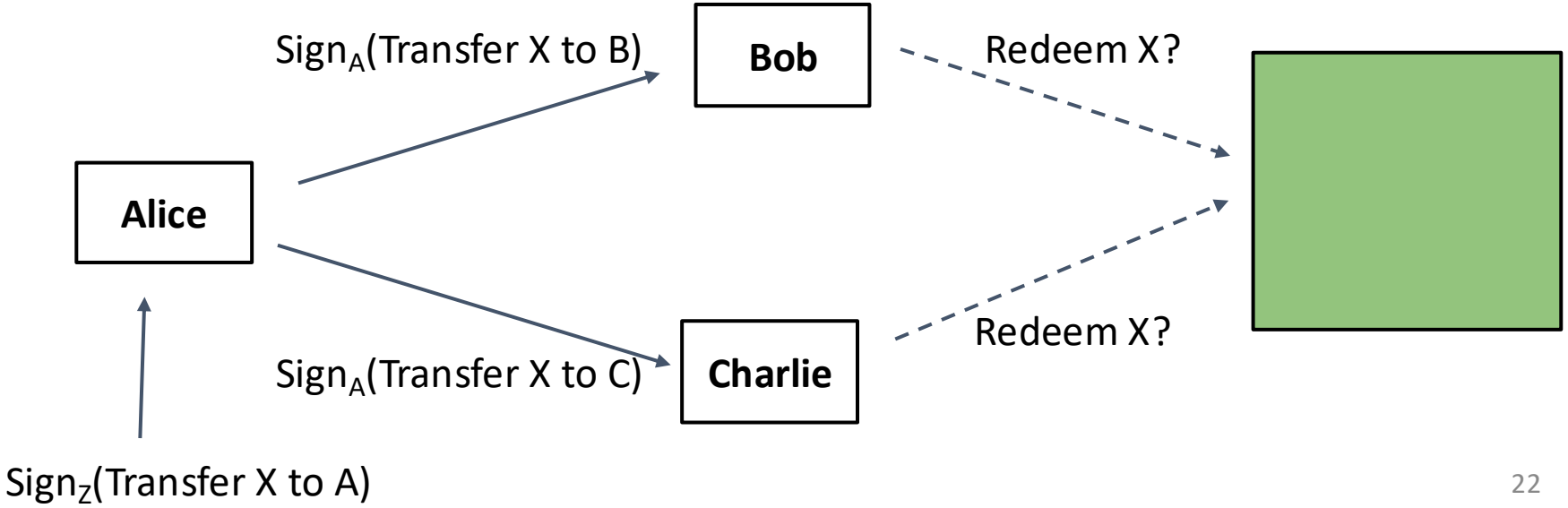


Double spending

- With traditional banking, avoid double spending by checking the account's balance
  - Centralized balance sheet
- ACH system for transferring money
  - Verify identity of the sender and recipient bank
  - Verify account amounts
  - Federal Reserve runs network

Double spending

- If you receive a Bitcoin, you can cryptographically verify the signatures
- But how can you verify that no double-spending took place?



Distributed ledger

- Distributed alternative to bank's centralized balance sheet
- Bitcoin participants engage in peer-to-peer protocol that shares transaction history
- They collectively build a log of every Bitcoin transaction, ever
- The log is built from *blocks*

- Everyone keeps a copy
- Append-only (no delete/edit)

Distributed ledger

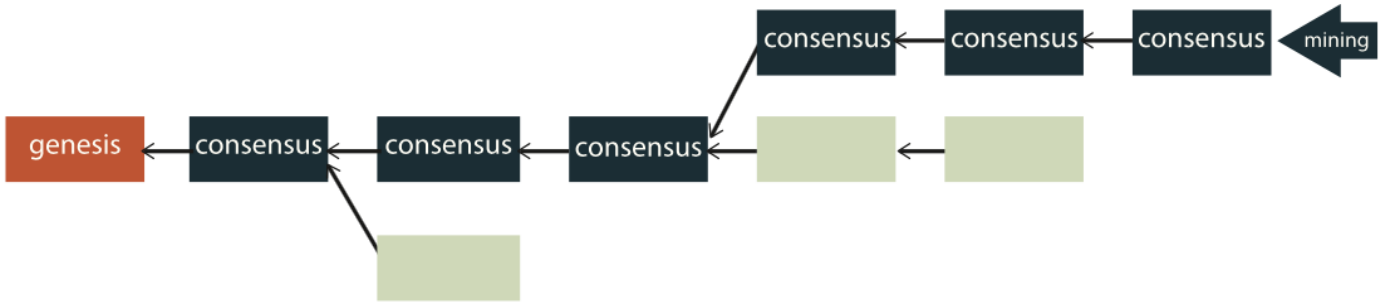
- Blockchain is essentially a shared database
- Multiple writers
- No trust required

Blocks

- Transactions grouped into blocks
- Each block has:
  - A sequence id
  - A timestamp
  - A nonce
  - Cryptographic hash of the previous block
  - Various metadata fields
  - Some actual transactions!

Blockchain

- Blocks connected in a *blockchain*
- There's always a backward hash link; eventually you can traverse it all the way back to the "genesis block"
- The set of blocks form a tree, with the genesis block at the root; possibly multiple branches



The blockchain in action

- Suppose Alice wants to give Bitcoins to Bob
- Alice creates and signs a transaction object that transfers the Bitcoin from herself to Bob
- She sends it to all peers in a peer-to-peer network
- The peers rebroadcast it, so soon, everyone knows about the transaction
- Any participant who also mines on the blockchain can now attempt to create a block that contains all the transactions known (including Alice's transfer)
- If this works, then the Alice-> Bob transfer is added to the global shared log of transactions

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Motivating Bitcoin miners

- Who runs the distributed ledger?
- *Bitcoin miners* are paid to operate the blockchain
  - Run servers
- Block creator is allowed to insert some **reward** Bitcoin transactions
- This is how miners earn Bitcoins, and how Bitcoins are created

Blockchain

- Bitcoin design decisions focus on building and maintaining a blockchain that is agreed-upon
- Rewards keep system honest
- Encourages consensus

Proof-of-work

- Goal: reject fake blocks
- Whoever solves a *proof-of-work* puzzle adds new block
  - Gets reward
- Proof-of-work is to compute, easy to validate

Proof-of-stake

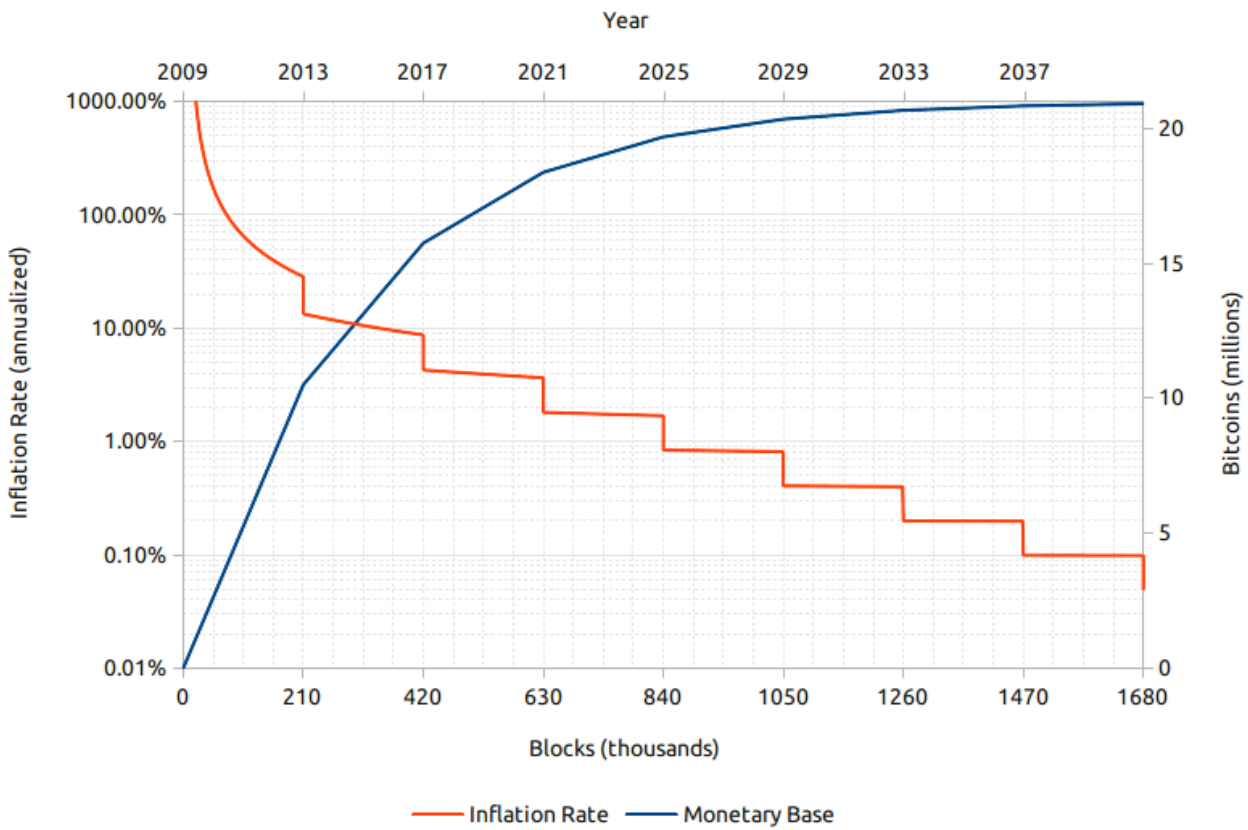
- Goal: reject fake blocks
- Whoever has the largest *proof-of-stake* adds new block
  - Gets reward
- Miners must hold and stake tokens for the privilege of earning transaction fees
  - Fake transactions and lose your staked coin
- Ethereum et al.

Blockchain

- Why give miners such strong incentives to formulate blocks?
  - Bring more mining power online
    - Reduces anyone's ability to forge an alternate history blockchain
  - You can start to ignore transactions if you control 51% of the hashing power

Bitcoin reward function

- Block reward halves every ~4 years



Energy consumption

- Current estimates: ~70 TWh / yr
- Enough to power ~7 million households
- Hardware is changing
  - Mining equipment: anything from normal CPU -> dedicated ASIC datacenter
- Power generation is changing
  - Coal, natural gas, wind, solar, hydroelectric

Source: https://digiconomist.net/bitcoin-energy-consumption

Energy consumption

- Why make the task so hard? Why proof-of-work?
- It caps the rate at which new Bitcoins can be added
- In other words, prevents runaway inflation

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Altcoins

- *Altcoins* are alternative cryptocurrencies, launched after Bitcoin
- Different blockchains offering different features

1	Ethereum	ETH	$ 3,573.43	$ 431.17B	-1.24%
2	XRP	XRP	$ 2.59	$ 150.07B	+6.90%
3	Tether USD	USDT	$ 1.00	$ 135.21B	+0.61%
4	Solana	SOL	$ 222.54	$ 105.86B	-1.21%
5	BNB	BNB	$ 639.69	$ 96.45B	+0.24%
6	Dogecoin	DOGE	$ 0.405	$ 59.68B	-3.14%
7	Cardano	ADA	$ 1.23	$ 46.14B	+11.27%
8	USDC	USDC	$ 1.00	$ 39.96B	+0.16%
9	Lido Staked Ether	stETH	$ 3,571.07	$ 34.16B	-1.17%
10	Avalanche	AVAX	$ 51.65	$ 21.28B	+6.68%

https://coinranking.com/coins/altcoin retrieved 12/03/2024

Cryptocurrency exchange

- A cryptocurrency exchange allows customers to
  - Bus/sell crypto
  - Convert to cash

- Examples: Poloniex, Bitfinex, Coinbase, Binance

BITFINEX

POLONIEX

Decentralized currency, centralized exchange

- Crypto exchanges are *centralized*

- Example: exchanging Bitcoin for Ethereum
  1. Alice sends Bitcoin to Bitfinex
  2. Bob sends Ethererum to Bitfinex
  3. Bitfinex makes the trade and takes a cut
  4. Bitfinex send Ethereum to Alice
  5. Bitfinex sends Bitcoin to Bob
- Both parties must trust Bitfinex

- What could possibly go wrong?

THE WALL STREET JOURNAL.

FINANCE | CURRENCIES

Sam Bankman-Fried Is Convicted of Fraud in FTX Collapse

New York federal jury finds crypto exchange's founder guilty of stealing billions of dollars from customers

Wall Street Journal, Nov. 2, 2023

Different blockchain features

- One reason to create a different cryptocurrency: different blockchain with different features
- Alternative proof of work
- Faster block times / larger block sizes
- Alternative transaction types
  - Computable “smart contract” transactions
  - Anonymous / Private transactions

Ethereum – Smart Contracts

- Altcoin (ETH)
- *Smart contract is a* computer program stored on the blockchain
- *Cannot be changed once deployed*
- *Automatically executes based on its code*

NFTs

- Non-Fungible Token (NFT): individually unique and provably scarce digital signature
  - Used to validate ownership
  - Analogous to a deed for digital goods
- Implemented with Ethereum smart contracts
  - Uses Ethereum blockchain
- E.g., a digital signature that says you own the JPG that hashes to 0x123ABC

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Reasons for cryptocurrencies

- Distrust in centralized banks or governments
- Predictable policy
  - Code and protocols instead of malleable laws
- Removing financial gatekeepers

Reasons against cryptocurrencies

- Should we trust anonymous people on the Internet?
- What about bugs in code?
- Does replacing financial gatekeepers with code mean that things will never change?
- Large power consumption causes environmental concerns
  - Maybe proof-of-stake will help?

Blockchain summary

- A distributed database with a shared protocol for multiple writers who don’t trust each other or a central authority
- Blockchain useful when
  - Database is shared with multiple writers
  - Users don’t trust each other
  - Users don’t trust a central authority
- Cryptocurrency is just one way to use a blockchain